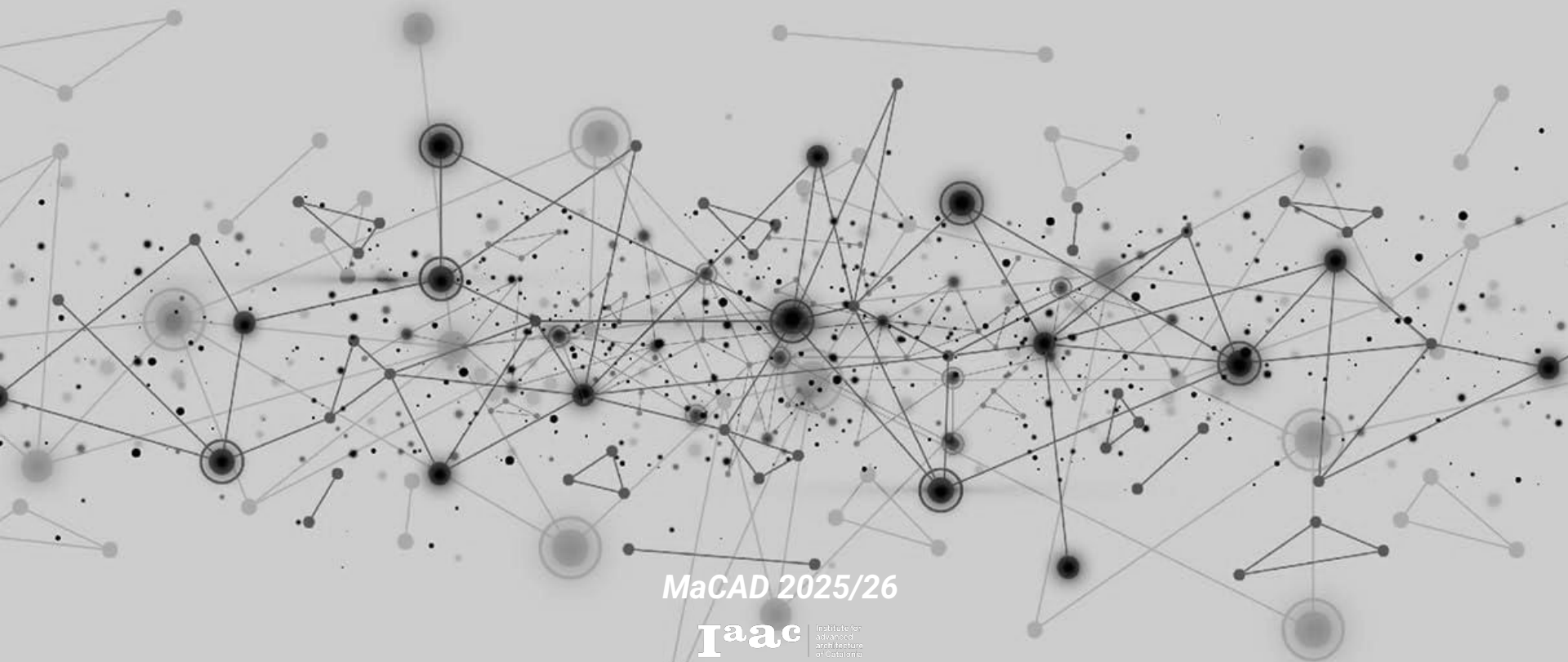


Graph Machine Learning Seminar

Professors: Wassim Jabi_ Olga Poletkina

Student_ Rania Chihaoui

Graph Machine Learning: Learn about the latest advancements in graph data to build robust machine learning models



MaCAD 2025/26

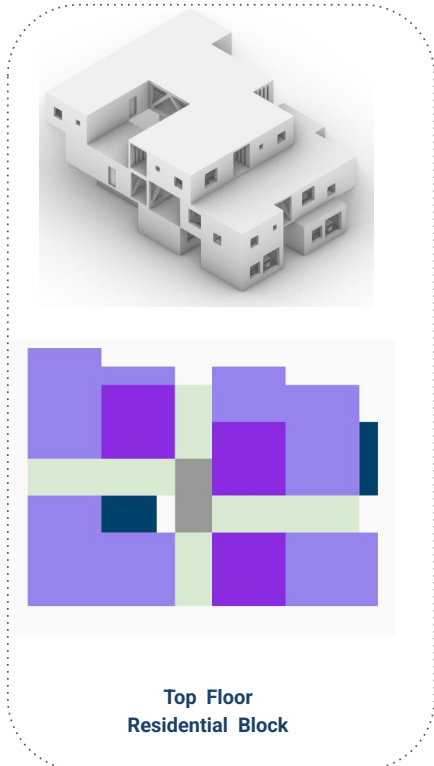
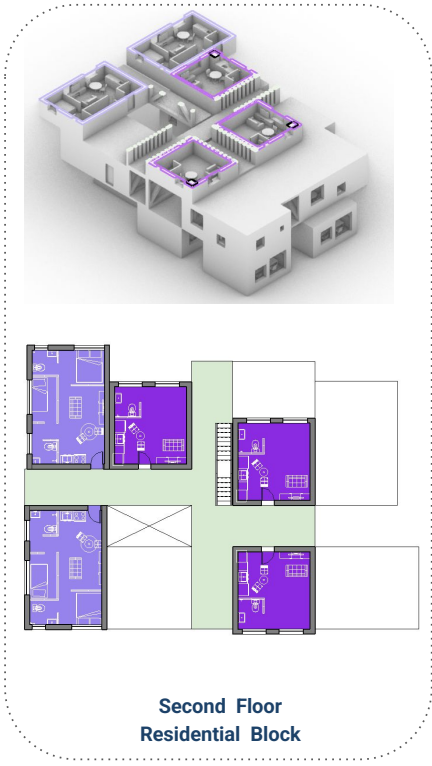
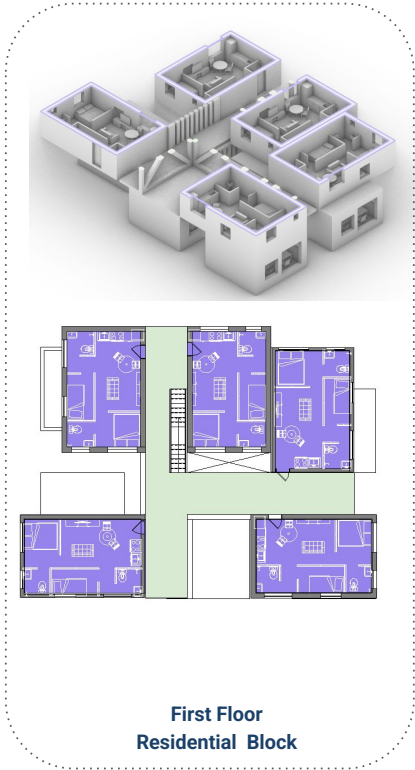
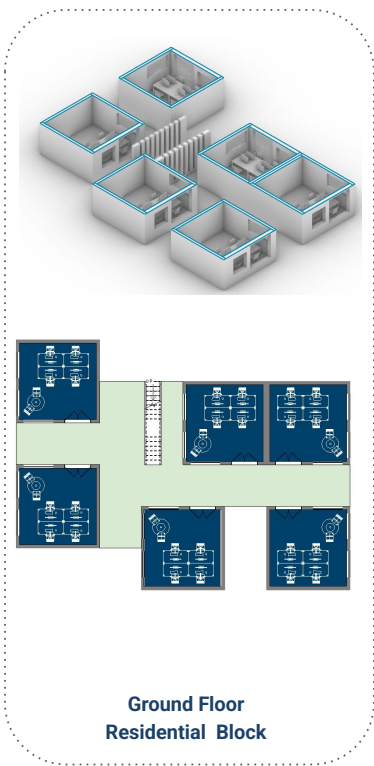
Iaac Institute for
Advanced
Architecture
of Catalonia

Assignment 01: Graph Generation from an Architectural Model

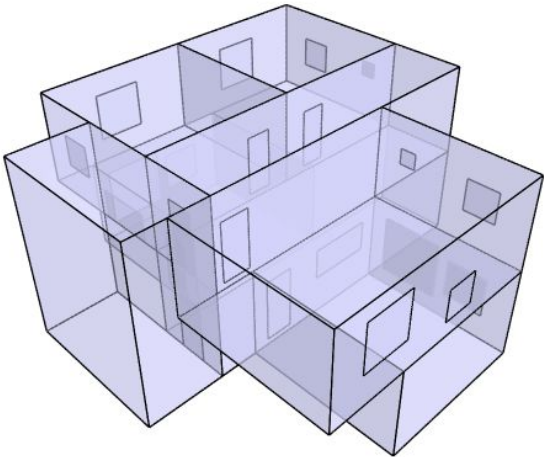
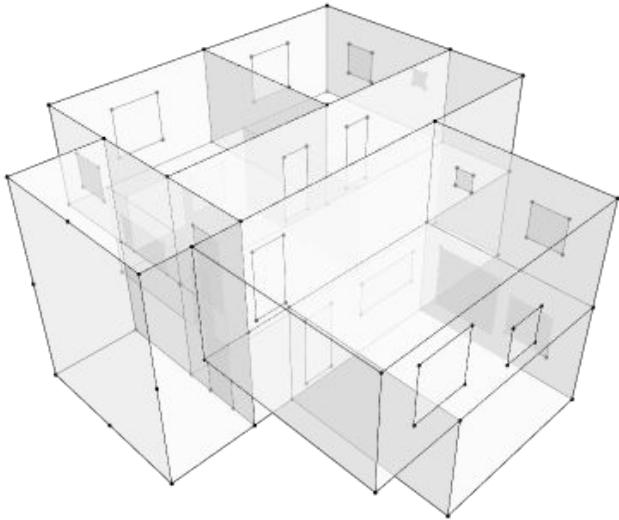
Project to analyse laac ACESD Studio project: Floating Grounds



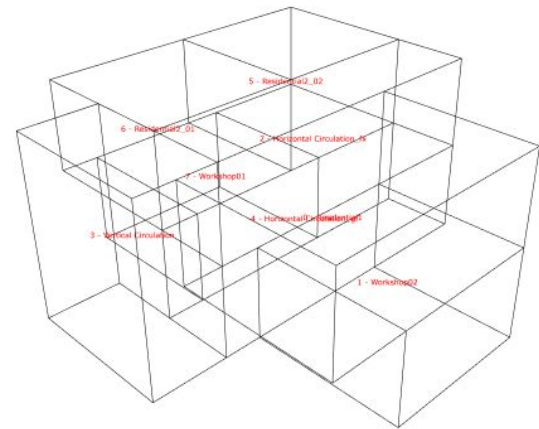
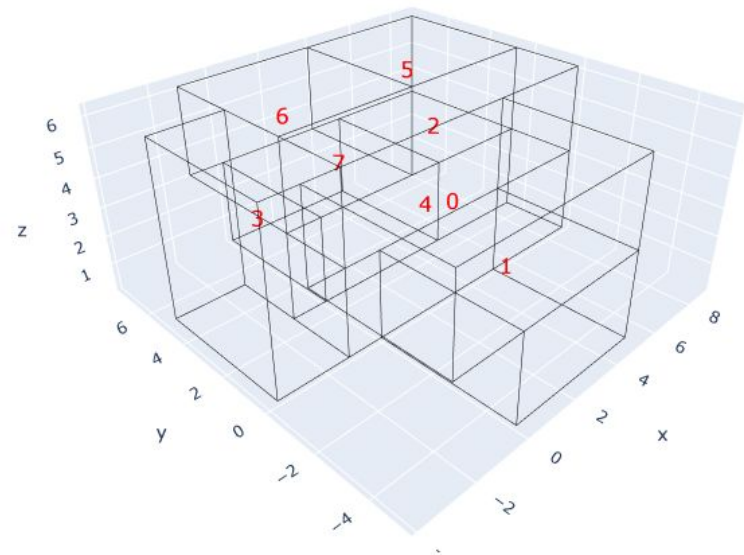
Residential Unit Example



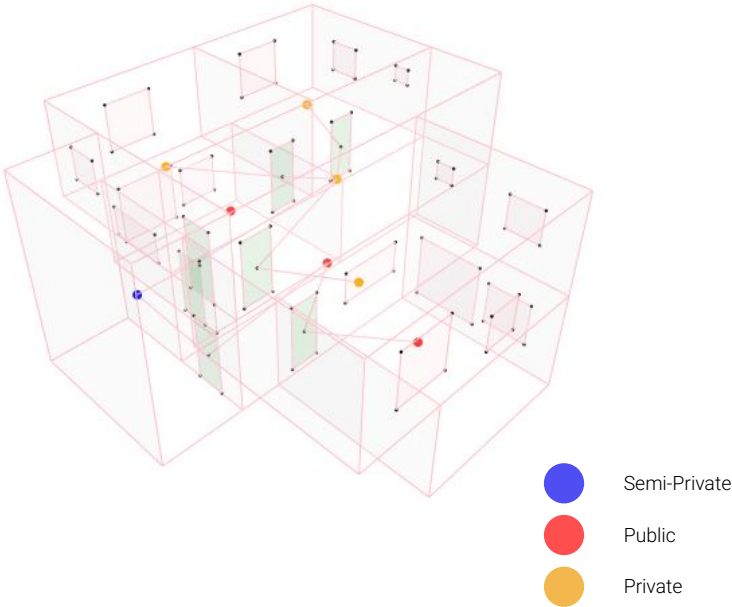
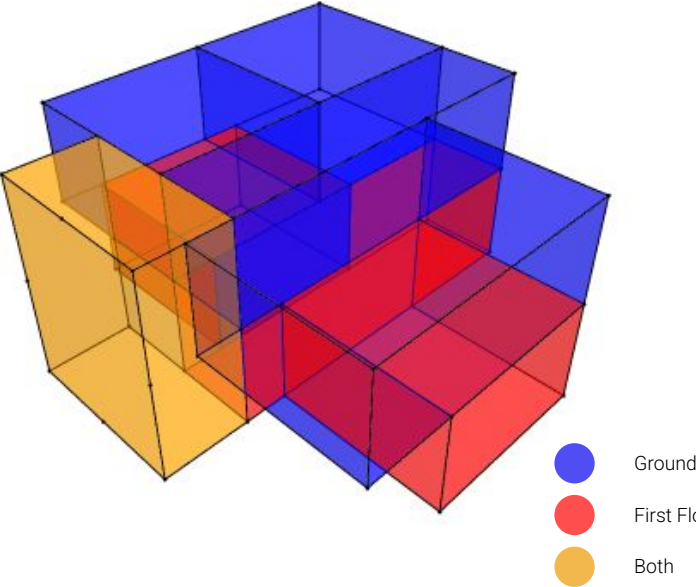
Simplified Graph Generation



Adding Label and Infi to the Rooms



Analysing The Rooms based on Floor Level and Privacy

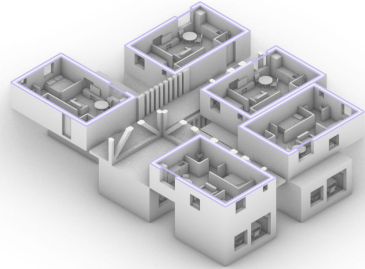


Assignment 02 : Spatial Analysis, Integration and Choice Metrics

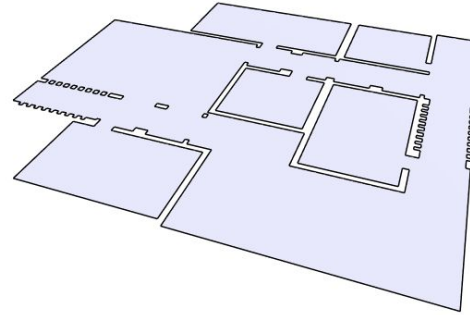
Project to analyse laac ACESD Studio project: Floating Grounds



Residential Unit Example



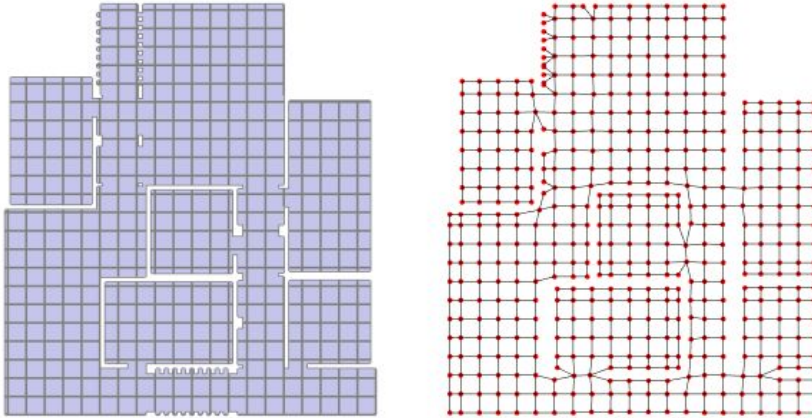
**First Floor
Residential Block**



This report evaluates a first-floor residential prototype through Space Syntax metrics to assess movement and privacy.

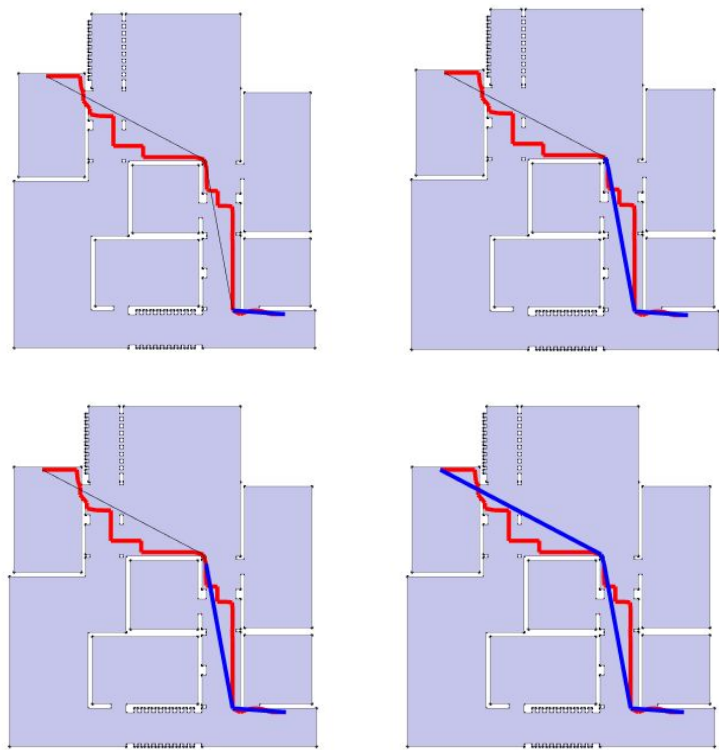
The results show a centralized layout where corridors serve as high-integration spines while private units remain appropriately segregated.

Methodology



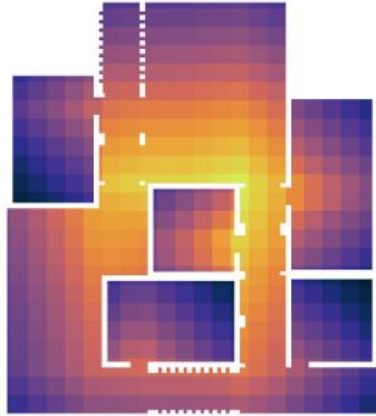
The floor plan was analyzed using a uniform grid and a "Straighten Wire" optimization to refine movement paths.

Methodology



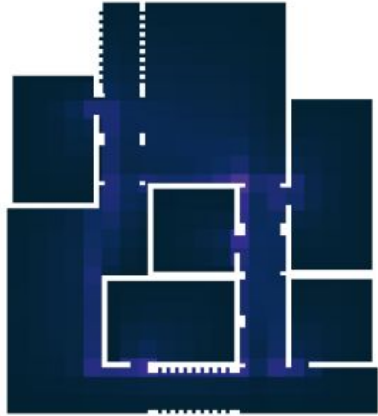
This process reduced the path length by 16.4%, confirming that the corridor geometry supports highly efficient circulation.

Integration (Closeness Centrality)



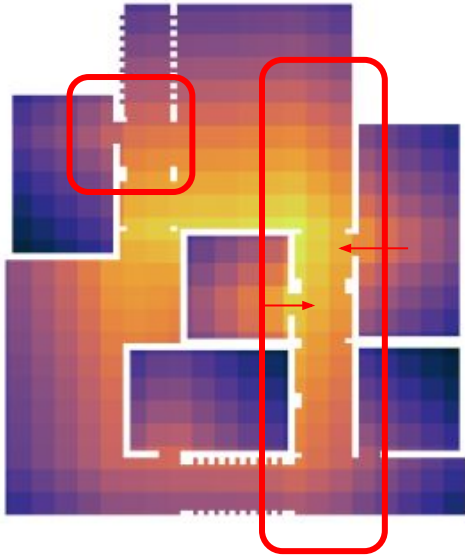
Highest integration values cluster at the T-shaped corridor junction, establishing it as the primary distribution hub for the entire floor. Private rooms remain in the low-integration "blue" zones, ensuring the acoustic buffering and seclusion necessary for residential living.

Choice (Betweenness Centrality)



Betweenness metrics identify the central threshold and right-side corridor as the most critical bridges for through-movement. These segments carry the highest traffic load and serve as the most direct routes for daily navigation and emergency egress.

Potential Weak Points and Recommendation



The high integration of the main corridor may feel institutional if apartment doors open directly onto the busiest zones without a buffer.

Additionally, the extreme segregation of corner units might lead to social isolation despite offering maximum quietness.

Adding recessed entryways for each apartment would create a necessary transitional threshold between the public corridor and private space. For the isolated corner units, maintaining their private status is ideal, but they could benefit from secondary light or air access to offset their topological depth.

Conclusion



The simulation confirms a functionally coherent plan that successfully balances a high-efficiency spine with private residential pockets. While the layout is mathematically sound for wayfinding, minor architectural refinements would better soften the transitions between social and private realms.

Limitations

Files too large

×



The following files are over 100MB. **If you commit these files, you will no longer be able to push this repository to GitHub.com.**

assets\obj\aggregation.obj

We recommend you avoid committing these files or use [Git LFS](#) to store large files on GitHub.

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